Lesson 9.1 - Z-Scores, Confidence Intervals, and Margin of Error

> **Video:** How To Find The Z Score, Confidence Interval, and Margin of Error for a Population Mean by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Confidence interval for μ (when σ is known)

$$\bar{x} - z\,\frac{\sigma}{\sqrt{n}} \;<\; \mu \;<\; \bar{x} + z\,\frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \bar{x} \pm \text{EBM}$$

Margin of error, also called the **error bound for the mean (EBM)**:

$$\text{EBM} = z\,\frac{\sigma}{\sqrt{n}}$$

Finding the critical z-value

The area to the **left** of the positive critical value z is

$$\text{left area} = \frac{\text{CL} + 1}{2}$$

where CL is the confidence level as a decimal. Look this area up in a z-table that gives **left** areas.

Confidence level	Left area	z
90%	0.95	1.645
95%	0.975	1.96

For 90%, the table has 0.94950 at $z = 1.64$ and 0.95053 at $z = 1.65$. 0.95 is about halfway between them, so average: $z = 1.645$.

Example 1 - physics test, 95% CI

$\sigma = 5.4$, $n = 50$, $\bar{x} = 79$

1. z: left area $= (0.95 + 1)/2 = 0.975 \rightarrow z = 1.96$
2. Standard error: $\dfrac{5.4}{\sqrt{50}} \approx 0.76368$
3. **(b) Margin of error:** $\text{EBM} = 1.96 \times 0.76368 \approx 1.4968$
4. **(a) Interval:** 79 ± 1.4968

$$77.5032 < \mu < 80.4968$$

We are 95% confident that the mean score of the whole class lies between about **77.50 and 80.50**.

Example 2 - chemistry test, 90% CI

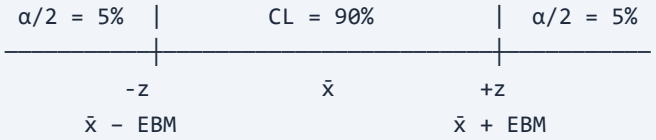
$\sigma = 6.5$, $n = 100$, $\bar{x} = 82$

1. z: left area $= (0.90 + 1)/2 = 0.95 \rightarrow z = 1.645$
2. **(b) Margin of error:** $\text{EBM} = 1.645 \times \dfrac{6.5}{\sqrt{100}} = 1.645 \times 0.65 = 1.06925$
3. **(a) Interval:** 82 ± 1.06925

$$80.93075 < \mu < 83.06925$$

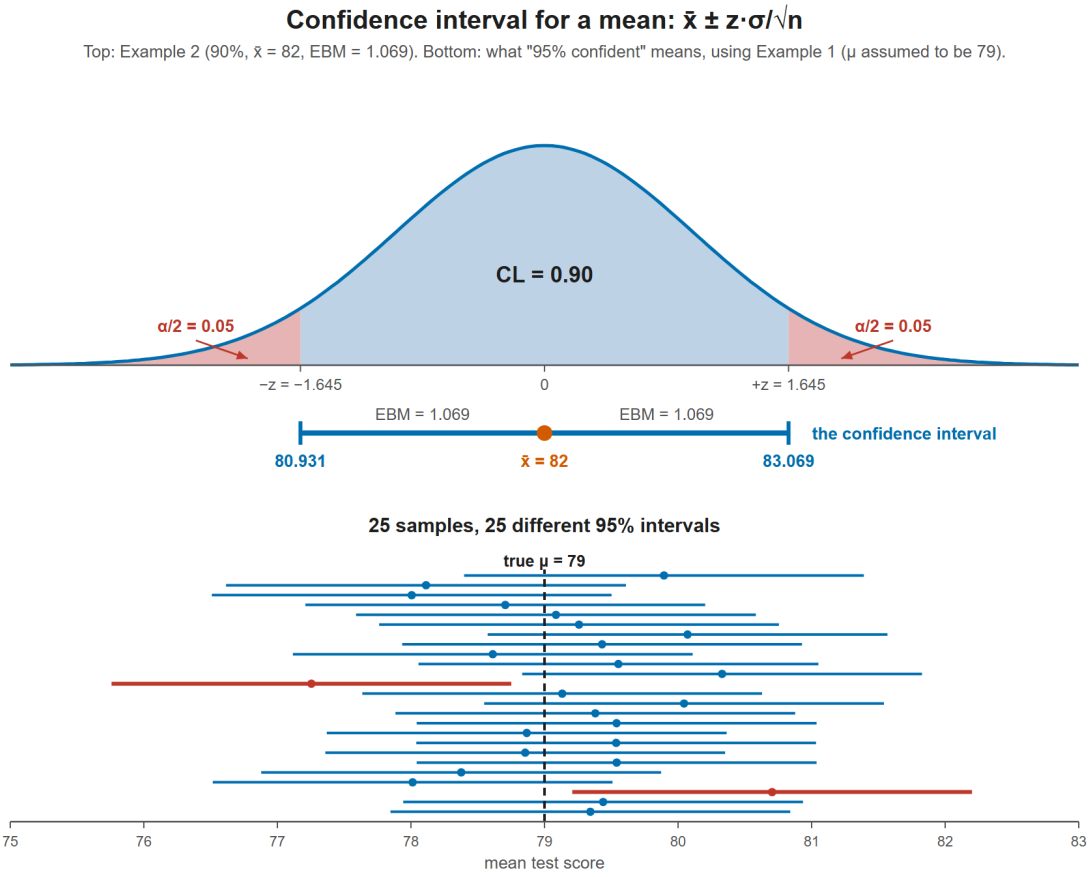
Confidence interval vs confidence level

Term	Meaning	Example
Confidence interval (CI)	The **range of values** $(\bar{x} - \text{EBM},\ \bar{x} + \text{EBM})$	80.93 to 83.07
Confidence level (CL)	The **area under the curve** between $-z$ and $+z$	0.90
α	The area **outside** the interval: $\alpha = 1 - \text{CL}$, split as $\alpha/2$ in each tail	0.10 → 0.05 per tail



Interpreting "95% confidence"

If we took many samples and built a 95% interval from each one, about **95% of those intervals would contain the true mean** μ, and about 5% would miss it. The video phrases this in terms of sample means landing in the interval; this is the more precise statement.



In this run 2 of the 25 intervals (red) miss μ. In the long run about 5% of 95% intervals miss.

Top: the confidence level is the middle area between â□□z and +z, and the interval reaches one EBM either side of xì□. Bottom: 25 simulated samples give 25 different intervals. Most of them cover the true mean (dashed line), but a few miss it, which is what "95% confident" means in the long run.

What changes the margin of error?

$$\text{EBM} = z \frac{\sigma}{\sqrt{n}}$$

Change	Effect on EBM	Why
Higher confidence level (90% → 95%)	**Larger**	z grows, so the interval gets wider
Larger sample size n	**Smaller**	$\sqrt{n}$ is in the denominator

To get a smaller margin of error, **lower the confidence level** or **increase the sample size**.

The EBM is half the width of the confidence interval.

Key takeaways

- CI = $\bar{x} \pm z\,\sigma/\sqrt{n}$, and EBM = $z\,\sigma/\sqrt{n}$.
- Critical z: look up the left area $(\mathrm{CL}+1)/2$. 90% → 1.645, 95% → 1.96.
- Higher confidence gives a wider interval. A bigger sample gives a narrower one.